

**The Hong Kong Polytechnic University**  
**Department of Land Surveying and Geo-Informatics**  
**LSGI3662 Field Scheme of Unmanned Aerial Vehicle (UAV) Photogrammetry and**  
**Laser Scanning**

## Survey Report

Subject Instructor: Mr. Nelson Chan

Group ■

[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]

Report Written: 24<sup>th</sup> December [REDACTED]

## Table of Contents

|                                                           |    |
|-----------------------------------------------------------|----|
| Abstract .....                                            | 2  |
| Objectives .....                                          | 3  |
| Methodologies.....                                        | 3  |
| a.    UAV Drones .....                                    | 3  |
| b.    Air LiDAR .....                                     | 4  |
| c.    GNSS RTK Survey .....                               | 5  |
| 3D Model Generation .....                                 | 6  |
| Accuracy check .....                                      | 7  |
| Achieved accuracy .....                                   | 7  |
| a.    Comparison with UAV data.....                       | 8  |
| b.    Comparison with GreenValley air LiDAR data.....     | 8  |
| c.    Comparison with DJI Zenmuse L1 air LiDAR data ..... | 9  |
| Accuracy summary .....                                    | 9  |
| a.    UAV.....                                            | 9  |
| b.    Air LiDAR .....                                     | 9  |
| Challenges Encountered.....                               | 10 |
| Self reflection and lessons learned.....                  | 11 |
| Recommendations .....                                     | 11 |
| Appendix.....                                             | 12 |
| A.    References.....                                     | 13 |
| B.    GNSS RTK Results.....                               | 13 |
| C.    Stations summaries for control points.....          | 13 |
| D.    Survey photos.....                                  | 16 |
| E.    Production .3sm file screenshots.....               | 23 |
| F.    Aerotriangulation quality reports .....             | 25 |

## Abstract

UAV and Air LiDAR surveys were conducted in Survey Area B, Tai Tong campsite, Yuen Long. The positions of the targets and control points were recorded with the GNSS RTK survey. This report first outlines the objectives and survey methodologies. This report delivers a 3D model generated using Bentley ContextCapture and airborne LiDAR, including three major sections: accuracy validation, methodology, and critical reflection.

In the accuracy validation part, the positional and vertical error from the UAV and Air LiDAR survey will be checked against the points from the RTK survey. In the methodology part, the technical, operational, and workflow aspects will be covered. This report also includes a critical reflection to identify the challenges encountered during the survey and provide suggestions for the survey.

## Objectives

This project aims to deliver 3D models generated by Context Capture and Air LiDAR, showing the accuracy of the point clouds, describing the methodologies in detail, and reflective analysis of the survey. We aim to apply the knowledge gained in lectures to the practical surveying tasks, while critically analyze the surveying procedures, identifying the challenges encountered during the survey and provide suggestions, ultimately aim to achieve learning from process.

## Methodologies

### a. UAV Drones

The UAV photogrammetry was used to capture the photos of the proximity of Survey Area B. The following steps summarizes the surveying procedures:

#### 1. Control point establishment

To recognize the targets for point cloud generation, April tags were placed on the control points and checkerboard targets were placed on the intended tie points with a wide sky window to avoid obstructions. The locations were recorded using GNSS equipment.

#### 2. Design flight plan



A flight plan with routes was designed to ensure the drone covers the whole survey area. The starting point should have a wide sky window to ensure ascension without obstruction. The height was set to 60m AGL and 70% overlaps to comply with the instructions and cover the area without the risks of crashing into obstacles.

The flight plan uses the Map Pilot Pro to design. It should be noted that the nadir flight plans were automatic and preset, while the oblique flight plans were hand operation manual operation to capture the structure of the building more accurately.

### 3. Role assignment

The flight plan was subdivided into 2 nadir sessions, and 4 oblique sessions. Since it was required to scan the Po Leung Kuk Jockey Club Leadership Training Base Building, the oblique sessions were focused on scanning sides of the building, with 3 oblique sessions at different levels, corresponding to 1<sup>st</sup>, 2<sup>nd</sup>, and 3<sup>rd</sup> floor respectively. The final oblique session flew high above the building to capture the top surface. The oblique sessions required manual operation, since flight plans were impractical in the environment, with many changes in angle and position, thus flight improvisation and adaptation was required. The nadir sessions cover the whole survey area B and the flight plans can be created before the survey.

Each member in the group will be the Chief Pilot and the Second Pilot at least once.

#### 3.1 Chief Pilot's responsibilities

- Overall direction and performance, command and control over SUA
- Keep flight incident and maintenance records
- Oversee flight safety

#### 3.2 Second Pilot's responsibilities

- Ensure good communication to the SUA
- Alert potential crashes
- Take over the Chief Pilot in case he was incapacitated
- Monitor UAV safety parameters, such as height, distance, battery level, and satellite count

### 4. UAV risk assessment and fail safe

The drone was placed on the starting point of the flight plan to adhere to the flight plan. Each safety component will be checked before taking off, such as the battery levels, signal strength, satellite counts, weather conditions. After initial take off, the Return to Home button will be pressed to ensure the fail-safe functionality was normal.

### 5. UAV survey

The drone will adhere to designated flight plans under automation for nadir photos and under manual control to capture oblique photos. The UAV flight parameters such as the battery levels will be monitored continuously by the Second Pilot, while the Chief Pilot will oversee the flight safety of the drone. After capturing all necessary photos, the drone will return to its starting point for landing.

#### b. Air LiDAR

The airborne LiDAR ("Air LiDAR") survey can compensate for the obstruction of trees and other objects, as well as shadowed areas. Since the UAV survey required obstruction-free environments to capture the ground features, and only capturing the topmost part of the terrain and buildings, LiDAR survey was used to cover the missed areas. When the signal hits the trees, it will return multiple signals. The equipment can resolve the multiple return issue and only keep the longest signal to ensure the signal only corresponds to the terrain instead of the canopy.

Furthermore, Air LiDAR uses a wider spectrum of EM wave, such as infrared light. This can provide more information than UAV photogrammetry, which relies on visible light, and can reflect signals even if the target blends with the surroundings in visible light.

The control points and target points were scanned by the LiDAR scanner, and the coordinates can be imported to Cyclone to geo-reference the production point cloud. It might require multiple attempts to fit the target points and reduce the error to below tolerance.

The weight of the Air LiDAR likely exceeded the threshold for A2 category SUA, and there were not enough Air LiDAR for each group to operate, thus it was operated by the instructors and exported to the students.

Below is an example of designing the flight plan using Map Pilot Pro:



Below shows the final result of generated 3D point cloud data in the RC



### c. GNSS RTK Survey

GNSS survey was conducted for geo-referencing the point cloud data. 29 physical ground features were selected for the survey. The surveying steps can be outlined as follows:

#### 1. Site reconnaissance

The rough locations of the physical features were selected during site reconnaissance. The physical features should be easily accessible for surveying with pole, and has a wide sky window to avoid obstructions.

## 2. Control point accuracy checking

To ensure the RTK accuracy was acceptable to start the survey, the GNSS survey first started with the 1 control stations with known coordinates. The expected error should be under 50mm. This process ensured that the control stations and the surveying equipment were reliable. After the GNSS survey for the control points and the tie points, the starting control station was also checked once to ensure that the survey results were consistent with the original coordinates.

## 3. GNSS surveying

Exact locations of 1 control stations, 6 control points, and 22 check points were selected with reference to the ones designated during site reconnaissance. During the surveying process, each point was awaited for at least 15 seconds to increase the time window for GNSS signals and increase the accuracy. This can increase the satellite redundancy of the survey results. The GDOP was also monitored to be less than 5 during the data collection. Each point was surveyed at least twice to further reduce the errors.

The following table summarizes the coordinates of all points:

| Point name        | Easting    | Northing   | Height |
|-------------------|------------|------------|--------|
| C1                | 821567.130 | 830138.748 | 91.408 |
| C2                | 821578.628 | 830313.298 | 71.811 |
| C3                | 821708.472 | 830305.872 | 62.274 |
| C4                | 821737.628 | 830268.596 | 61.628 |
| C5                | 821771.108 | 830240.738 | 61.440 |
| C6                | 821727.402 | 830220.114 | 70.533 |
| CP2 (start check) | 821291.997 | 830233.929 | 69.017 |
| CP2 (end check)   | 821291.999 | 830233.928 | 69.015 |
| T01               | 821569.072 | 830134.370 | 91.343 |
| T02               | 821624.408 | 830163.271 | 91.401 |
| T03               | 821631.015 | 830167.642 | 91.370 |
| T04               | 821587.166 | 830153.109 | 91.043 |
| T05               | 821579.589 | 830146.943 | 91.065 |
| T06               | 821564.534 | 830137.204 | 91.429 |
| T07               | 821634.624 | 830305.018 | 70.842 |
| T08               | 821674.139 | 830277.554 | 70.997 |
| T09               | 821724.544 | 830218.860 | 70.453 |
| T10               | 821710.790 | 830306.292 | 61.796 |
| T11               | 821713.069 | 830297.904 | 62.758 |
| T12               | 821716.616 | 830289.928 | 62.098 |
| T13               | 821764.096 | 830254.397 | 61.418 |
| T14               | 821753.315 | 830244.445 | 61.515 |
| T15               | 821749.059 | 830256.707 | 61.621 |
| T16               | 821752.032 | 830259.594 | 61.572 |

|     |            |            |        |
|-----|------------|------------|--------|
| T17 | 821746.321 | 830265.570 | 61.583 |
| T18 | 821743.812 | 830263.133 | 61.617 |
| T19 | 821741.181 | 830271.966 | 61.566 |
| T20 | 821750.521 | 830269.471 | 61.539 |
| T21 | 821731.966 | 830220.379 | 70.765 |

### 3D Model Generation

After collecting the data with the UAV and air LiDAR surveys, the raw data was exported to the corresponding software. For UAV survey, the aerial photos were exported to a new project in iTwin Capture Modeler, and for LiDAR survey, the point clouds were exported directly using Cyclone Register 360. The post processing procedures can be summarized as follows:

#### 1. Import the control point coordinates into iTwin Capture Modeler

In this project, 6 control points were used. They were be imported in the Survey Points tab, under the Hong Kong 1980 Grid Systems. The positional accuracy was changed to 50mm to match with the accuracy of the RTK.

#### 2. Aerotriangulation submission

During this process, the engine automatically merges the photos by recognizing the April tags in the photos and overlaps to form a 3D point cloud.

#### 3. Add check points and redo aerotriangulation

The check points can be added and the aerotriangulation can be redone again to make further adjustments. A quality report can be generated to view the accuracy of the control points. The color should be green, indicating the error was within one pixel.

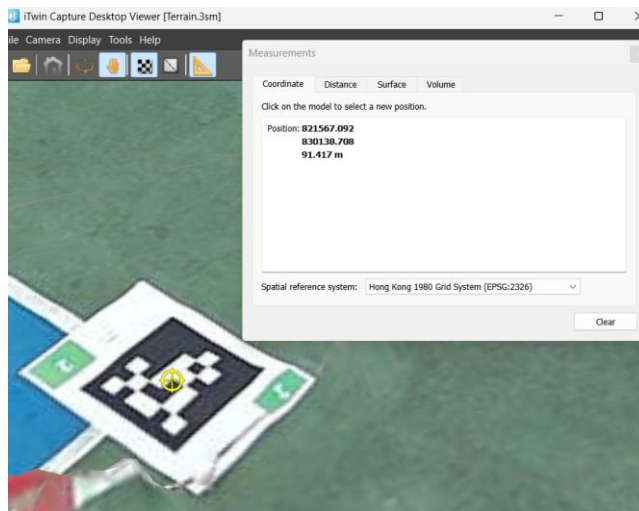
#### 4. Model finalization

The final quality report can be generated to retrieve the final adjusted coordinates of the control points and check points. They can be used to check against the RTK survey to view their positional and vertical accuracies. The 3D models and project can also be exported, such as .obj file and .3sm project files.

During the aero triangulation and merging process, technical problems arose and we could not combine the building point clouds with the terrain point clouds. Therefore, the point clouds were processed separately and produced into 2 separate .3sm files, namely Building.3sm and Terrain.3sm.

### Accuracy check

The coordinates from the Air LiDAR point clouds and the 3D mesh can be compared against the coordinates measured by RTK. This can check the accuracy of the UAV and Air LiDAR surveys. At least 20 check points, including the control points used for UAV survey, were used to ensure the points cover the whole survey area and reduce regional biases.



iTwin Capture Desktop Viewer was used to measure the coordinates of the .3sm production mesh files, as shown on the left, while Potree Viewer was used to measure the coordinates of the point clouds produced by the Air LiDAR.

## Achieved accuracy

The point cloud data produced by ContextCapture and Air LiDAR contain the adjusted positions of the known points. The error can be obtained by subtracting the adjusted points with the coordinates surveyed by RTK. The positional and vertical accuracy can be obtained by finding the standard deviation of positional and vertical errors.

Below are the spreadsheets for accuracy calculations for UAV and Air LiDAR surveys against the RTK survey results:

### a. Comparison with UAV data

| Name | GNSS RTK   |            |        | Context Capture Data |            |        | Difference |        |        |         | RMSE (accuracy) |       | SD of errors (precision) |        |
|------|------------|------------|--------|----------------------|------------|--------|------------|--------|--------|---------|-----------------|-------|--------------------------|--------|
|      | X          | Y          | Z      | X                    | Y          | Z      | dx         | dy     | dZ     | dHoriz. | dHoriz.         | dZ    | σ Horiz.                 | σZ     |
| C1   | 821567.130 | 830138.748 | 91.408 | 821567.094           | 830138.708 | 91.417 | -0.036     | -0.040 | 0.009  | 0.054   | 0.0502          | 0.092 | 0.0236                   | 0.0797 |
| C2   | 821578.628 | 830313.298 | 71.811 | 821578.686           | 830313.256 | 71.764 | 0.058      | -0.042 | -0.047 | 0.072   |                 |       |                          |        |
| C3   | 821708.472 | 830305.872 | 62.274 | 821708.510           | 830305.894 | 62.268 | 0.038      | 0.022  | -0.006 | 0.044   |                 |       |                          |        |
| C4   | 821737.628 | 830268.596 | 61.628 | 821737.633           | 830268.625 | 61.665 | 0.005      | 0.029  | 0.037  | 0.029   |                 |       |                          |        |
| C5   | 821771.108 | 830240.738 | 61.440 | 821771.102           | 830240.749 | 61.461 | -0.006     | 0.011  | 0.021  | 0.013   |                 |       |                          |        |
| C6   | 821727.402 | 830220.114 | 70.533 | 821727.345           | 830220.084 | 70.616 | -0.057     | -0.030 | 0.083  | 0.064   |                 |       |                          |        |
| T01  | 821569.072 | 830134.370 | 91.343 | 821569.056           | 830134.359 | 91.345 | -0.016     | -0.011 | 0.002  | 0.019   |                 |       |                          |        |
| T02  | 821624.408 | 830163.271 | 91.401 | 821624.463           | 830163.228 | 91.382 | 0.055      | -0.043 | -0.019 | 0.070   |                 |       |                          |        |
| T03  | 821631.015 | 830167.642 | 91.370 | 821631.082           | 830167.583 | 91.389 | 0.067      | -0.059 | 0.019  | 0.089   |                 |       |                          |        |
| T04  | 821587.166 | 830153.109 | 91.043 | 821587.154           | 830153.113 | 91.054 | -0.012     | 0.004  | 0.011  | 0.013   |                 |       |                          |        |
| T05  | 821579.589 | 830146.943 | 91.065 | 821579.532           | 830146.918 | 91.155 | -0.057     | -0.025 | 0.090  | 0.062   |                 |       |                          |        |
| T06  | 821564.534 | 830137.204 | 91.429 | 821564.496           | 830137.170 | 91.419 | -0.038     | -0.034 | -0.010 | 0.051   |                 |       |                          |        |
| T07  | 821634.624 | 830305.018 | 70.842 | 821634.623           | 830305.005 | 71.080 | -0.001     | -0.013 | 0.238  | 0.013   |                 |       |                          |        |
| T08  | 821674.139 | 830277.554 | 70.997 | 821674.120           | 830277.503 | 71.252 | -0.019     | -0.051 | 0.255  | 0.054   |                 |       |                          |        |
| T09  | 821724.544 | 830218.860 | 70.453 | 821724.467           | 830218.842 | 70.482 | -0.077     | -0.018 | 0.029  | 0.079   |                 |       |                          |        |
| T10  | 821710.790 | 830306.292 | 61.796 | 821710.830           | 830306.321 | 61.791 | 0.040      | 0.029  | -0.005 | 0.049   |                 |       |                          |        |
| T11  | 821713.069 | 830297.904 | 62.758 | 821713.105           | 830297.933 | 62.849 | 0.036      | 0.029  | 0.091  | 0.046   |                 |       |                          |        |
| T12  | 821716.616 | 830289.928 | 62.098 | 821716.649           | 830289.964 | 62.088 | 0.033      | 0.036  | -0.010 | 0.049   |                 |       |                          |        |
| T13  | 821764.096 | 830254.397 | 61.418 | 821764.089           | 830254.392 | 61.519 | -0.007     | -0.005 | 0.101  | 0.009   |                 |       |                          |        |
| T14  | 821753.315 | 830244.445 | 61.515 | 821753.317           | 830244.482 | 61.429 | 0.002      | 0.037  | -0.086 | 0.037   |                 |       |                          |        |
| T15  | 821749.059 | 830256.707 | 61.621 | 821749.072           | 830256.746 | 61.579 | 0.013      | 0.039  | -0.042 | 0.041   |                 |       |                          |        |
| T16  | 821752.032 | 830259.594 | 61.572 | 821752.035           | 830259.629 | 61.596 | 0.003      | 0.035  | 0.024  | 0.035   |                 |       |                          |        |
| T17  | 821746.321 | 830265.570 | 61.583 | 821746.320           | 830265.590 | 61.696 | -0.001     | 0.020  | 0.113  | 0.020   |                 |       |                          |        |
| T18  | 821743.812 | 830263.133 | 61.617 | 821743.813           | 830263.178 | 61.703 | 0.001      | 0.045  | 0.086  | 0.045   |                 |       |                          |        |
| T19  | 821741.181 | 830271.966 | 61.566 | 821741.179           | 830271.963 | 61.705 | -0.002     | -0.003 | 0.139  | 0.004   |                 |       |                          |        |
| T20  | 821750.521 | 830269.471 | 61.539 | 821750.520           | 830269.533 | 61.608 | -0.001     | 0.062  | 0.069  | 0.062   |                 |       |                          |        |
| T21  | 821731.966 | 830220.379 | 70.765 | 821731.890           | 830220.366 | 70.880 | -0.076     | -0.013 | 0.115  | 0.077   |                 |       |                          |        |

Root mean square error (RMSE) of horizontal measurements = 0.0502m

RMSE of vertical measurements = 0.0920m

Standard deviation (SD) of horizontal measurement errors = 0.0236m

SD of vertical measurement errors = 0.0796m

### b. Comparison with GreenValley Air LiDAR data

| Name | GNSS RTK   |            |        | GreenValley AirLiDAR Data |            |        | Difference |        |        |         | RMSE (accuracy) |        | SD of errors (precision) |        |
|------|------------|------------|--------|---------------------------|------------|--------|------------|--------|--------|---------|-----------------|--------|--------------------------|--------|
|      | X          | Y          | Z      | X                         | Y          | Z      | dX         | dY     | dZ     | dHoriz. | dHoriz.         | dZ     | σ Horiz.                 | σZ     |
| C1   | 821567.130 | 830138.748 | 91.408 | 821567.120                | 830138.720 | 91.380 | -0.010     | -0.028 | -0.028 | 0.030   | 0.1485          | 0.0532 | 0.0916                   | 0.0463 |
| C2   | 821578.628 | 830313.298 | 71.811 | 821578.720                | 830313.400 | 71.830 | 0.092      | 0.102  | 0.019  | 0.137   |                 |        |                          |        |
| C3   | 821708.472 | 830305.872 | 62.274 | 821708.410                | 830305.900 | 62.250 | -0.062     | 0.028  | -0.024 | 0.068   |                 |        |                          |        |
| C4   | 821737.628 | 830268.596 | 61.628 | 821737.590                | 830268.600 | 61.610 | -0.038     | 0.004  | -0.018 | 0.038   |                 |        |                          |        |
| C5   | 821771.108 | 830240.738 | 61.440 | 821771.110                | 830240.760 | 61.430 | 0.002      | 0.022  | -0.010 | 0.022   |                 |        |                          |        |
| C6   | 821727.402 | 830220.114 | 70.533 | 821727.280                | 830220.150 | 70.510 | -0.122     | 0.036  | -0.023 | 0.127   |                 |        |                          |        |
| T01  | 821569.072 | 830134.370 | 91.343 | 821569.000                | 830134.320 | 91.320 | -0.072     | -0.050 | -0.023 | 0.088   |                 |        |                          |        |
| T02  | 821624.408 | 830163.271 | 91.401 | 821624.440                | 830163.280 | 91.310 | 0.032      | 0.009  | -0.091 | 0.033   |                 |        |                          |        |
| T03  | 821631.015 | 830167.642 | 91.370 | 821631.010                | 830167.690 | 91.150 | -0.005     | 0.048  | -0.220 | 0.048   |                 |        |                          |        |
| T04  | 821587.166 | 830153.109 | 91.043 | 821587.040                | 830153.180 | 91.000 | -0.126     | 0.071  | -0.043 | 0.145   |                 |        |                          |        |
| T05  | 821579.589 | 830146.943 | 91.065 | 821579.430                | 830146.820 | 91.100 | -0.159     | -0.123 | 0.035  | 0.201   |                 |        |                          |        |
| T06  | 821564.534 | 830137.204 | 91.429 | 821564.530                | 830137.180 | 91.420 | -0.004     | -0.024 | -0.009 | 0.024   |                 |        |                          |        |
| T07  | 821634.624 | 830305.018 | 70.842 | 821634.390                | 830305.310 | 70.810 | -0.234     | 0.292  | -0.032 | 0.374   |                 |        |                          |        |
| T08  | 821674.139 | 830277.554 | 70.997 | 821674.110                | 830277.390 | 70.950 | -0.029     | -0.164 | -0.047 | 0.167   |                 |        |                          |        |
| T09  | 821724.544 | 830218.860 | 70.453 | 821724.360                | 830218.860 | 70.400 | -0.184     | 0.000  | -0.053 | 0.184   |                 |        |                          |        |
| T10  | 821710.790 | 830306.292 | 61.796 | 821710.660                | 830306.050 | 61.810 | -0.130     | -0.242 | 0.014  | 0.275   |                 |        |                          |        |
| T11  | 821713.069 | 830297.904 | 62.758 | 821713.120                | 830297.900 | 62.750 | 0.051      | -0.004 | -0.008 | 0.051   |                 |        |                          |        |
| T12  | 821716.616 | 830289.928 | 62.098 | 821716.610                | 830289.890 | 62.070 | -0.006     | -0.038 | -0.028 | 0.038   |                 |        |                          |        |
| T13  | 821764.096 | 830254.397 | 61.418 | 821764.220                | 830254.300 | 61.400 | 0.124      | -0.097 | -0.018 | 0.157   |                 |        |                          |        |
| T14  | 821753.315 | 830244.445 | 61.515 | 821753.530                | 830244.570 | 61.510 | 0.215      | 0.125  | -0.005 | 0.249   |                 |        |                          |        |
| T15  | 821749.059 | 830256.707 | 61.621 | 821749.280                | 830256.740 | 61.600 | 0.221      | 0.033  | -0.021 | 0.223   |                 |        |                          |        |
| T16  | 821752.032 | 830259.594 | 61.572 | 821752.040                | 830259.590 | 61.560 | 0.008      | -0.004 | -0.012 | 0.009   |                 |        |                          |        |
| T17  | 821746.321 | 830265.570 | 61.583 | 821746.490                | 830265.570 | 61.580 | 0.169      | 0.000  | -0.003 | 0.169   |                 |        |                          |        |
| T18  | 821743.812 | 830263.133 | 61.617 | 821743.930                | 830263.110 | 61.580 | 0.118      | -0.023 | -0.037 | 0.120   |                 |        |                          |        |
| T19  | 821741.181 | 830271.966 | 61.566 | 821741.260                | 830271.930 | 61.580 | 0.079      | -0.036 | 0.014  | 0.087   |                 |        |                          |        |
| T20  | 821750.521 | 830269.471 | 61.539 | 821750.520                | 830269.450 | 61.480 | -0.001     | -0.021 | -0.059 | 0.021   |                 |        |                          |        |
| T21  | 821731.966 | 830220.379 | 70.765 | 821732.070                | 830220.400 | 70.750 | 0.104      | 0.021  | -0.015 | 0.106   |                 |        |                          |        |

RMSE of horizontal measurements = 0.1485m

RMSE of vertical measurements = 0.0532m

SD of horizontal measurement errors = 0.0916 m

SD of vertical measurement errors = 0.0463 m

### c. Comparison with DJI Zenmuse L1 Air LiDAR data

| Name | GNSS RTK   |            |        | DJI Zenmuse L1 AirLiDAR Data |            |        | Difference |        |        |         | RMSE (accuracy) |       | SD of errors (precision) |        |
|------|------------|------------|--------|------------------------------|------------|--------|------------|--------|--------|---------|-----------------|-------|--------------------------|--------|
|      | X          | Y          | Z      | X                            | Y          | Z      | dX         | dY     | dZ     | dHoriz. | dHoriz.         | dZ    | σ Horiz.                 | σZ     |
| C1   | 821567.130 | 830138.748 | 91.408 | 821567.150                   | 830138.730 | 87.850 | 0.020      | -0.018 | -3.558 | 0.027   | 0.0792          | 3.587 | 0.0433                   | 0.0322 |
| C2   | 821578.628 | 830313.298 | 71.811 | 821578.630                   | 830313.270 | 68.250 | 0.002      | -0.028 | -3.561 | 0.028   |                 |       |                          |        |
| C3   | 821708.472 | 830305.872 | 62.274 | 821708.490                   | 830305.870 | 58.670 | 0.018      | -0.002 | -3.604 | 0.018   |                 |       |                          |        |
| C4   | 821737.628 | 830268.596 | 61.628 | 821737.620                   | 830268.630 | 57.990 | -0.008     | 0.034  | -3.638 | 0.035   |                 |       |                          |        |
| C5   | 821771.108 | 830240.738 | 61.440 | 821771.100                   | 830240.600 | 57.860 | -0.008     | -0.138 | -3.580 | 0.138   |                 |       |                          |        |
| C6   | 821727.402 | 830220.114 | 70.533 | 821727.450                   | 830220.090 | 66.940 | 0.048      | -0.024 | -3.593 | 0.054   |                 |       |                          |        |
| T01  | 821569.072 | 830134.370 | 91.343 | 821569.050                   | 830134.310 | 87.770 | -0.022     | -0.060 | -3.573 | 0.064   |                 |       |                          |        |
| T02  | 821624.408 | 830163.271 | 91.401 | 821624.470                   | 830163.200 | 87.800 | 0.062      | -0.071 | -3.601 | 0.094   |                 |       |                          |        |
| T03  | 821631.015 | 830167.642 | 91.370 | 821631.100                   | 830167.590 | 87.780 | 0.085      | -0.052 | -3.590 | 0.100   |                 |       |                          |        |
| T04  | 821587.166 | 830153.109 | 91.043 | 821587.190                   | 830153.080 | 87.470 | 0.024      | -0.029 | -3.573 | 0.038   |                 |       |                          |        |
| T05  | 821579.589 | 830146.943 | 91.065 | 821579.630                   | 830146.910 | 87.490 | 0.041      | -0.033 | -3.575 | 0.053   |                 |       |                          |        |
| T06  | 821564.534 | 830137.204 | 91.429 | 821564.450                   | 830137.210 | 87.850 | -0.084     | 0.006  | -3.579 | 0.084   |                 |       |                          |        |
| T07  | 821634.624 | 830305.018 | 70.842 | 821634.580                   | 830304.990 | 67.230 | -0.044     | -0.028 | -3.612 | 0.052   |                 |       |                          |        |
| T08  | 821674.139 | 830277.554 | 70.997 | 821674.150                   | 830277.600 | 67.390 | 0.011      | 0.046  | -3.607 | 0.047   |                 |       |                          |        |
| T09  | 821724.544 | 830218.860 | 70.453 | 821724.670                   | 830218.720 | 66.870 | 0.126      | -0.140 | -3.583 | 0.188   |                 |       |                          |        |
| T10  | 821710.790 | 830306.292 | 61.796 | 821710.690                   | 830306.330 | 58.100 | -0.100     | 0.038  | -3.696 | 0.107   |                 |       |                          |        |
| T11  | 821713.069 | 830297.904 | 62.758 | 821712.960                   | 830297.990 | 59.160 | -0.109     | 0.086  | -3.598 | 0.139   |                 |       |                          |        |
| T12  | 821716.616 | 830289.928 | 62.098 | 821716.610                   | 830289.940 | 58.510 | -0.006     | 0.012  | -3.588 | 0.013   |                 |       |                          |        |
| T13  | 821764.096 | 830254.397 | 61.418 | 821764.020                   | 830254.480 | 57.840 | -0.076     | 0.083  | -3.578 | 0.113   |                 |       |                          |        |
| T14  | 821753.315 | 830244.445 | 61.515 | 821753.330                   | 830244.460 | 57.960 | 0.015      | 0.015  | -3.555 | 0.021   |                 |       |                          |        |
| T15  | 821749.059 | 830256.707 | 61.621 | 821749.05                    | 830256.740 | 58.040 | -0.009     | 0.033  | -3.581 | 0.034   |                 |       |                          |        |
| T16  | 821752.032 | 830259.594 | 61.572 | 821751.960                   | 830259.610 | 58.010 | -0.072     | 0.016  | -3.562 | 0.074   |                 |       |                          |        |
| T17  | 821746.321 | 830265.570 | 61.583 | 821746.310                   | 830265.600 | 58.020 | -0.011     | 0.030  | -3.563 | 0.032   |                 |       |                          |        |
| T18  | 821743.812 | 830263.133 | 61.617 | 821743.760                   | 830263.170 | 58.060 | -0.052     | 0.037  | -3.557 | 0.064   |                 |       |                          |        |
| T19  | 821741.181 | 830271.966 | 61.566 | 821741.200                   | 830271.950 | 58.040 | 0.019      | -0.016 | -3.526 | 0.025   |                 |       |                          |        |
| T20  | 821750.521 | 830269.471 | 61.539 | 821750.590                   | 830269.440 | 57.940 | 0.069      | -0.031 | -3.599 | 0.076   |                 |       |                          |        |
| T21  | 821731.966 | 830220.379 | 70.765 | 821732.020                   | 830220.310 | 67.140 | 0.054      | -0.069 | -3.625 | 0.088   |                 |       |                          |        |

RMSE of horizontal measurements = 0.0792m

RMSE of vertical measurements = 3.5870m

SD of horizontal measurement errors = 0.0433 m

SD of vertical measurement errors = 0.0322 m

### Accuracy summary

### **a. UAV**

From the table, it appears that for horizontal measurements, both RMSE and SD of errors were less than vertical. This shows that for UAV photogrammetry, horizontal positions are likely more accurate and precise. This consistency is also reflected in the sharp imagery in the final production. However, the vertical accuracy can vary; sometimes the error was as large as 25cm. This is likely due to area obstruction and the lack of photos, the terrain level can differ significantly. For instance, at checkpoints T13 and T14, the sides of the pavement were covered by dense tree canopy, which could reduce the vertical accuracy.

Overall, this shows that UAV can be a reliable equipment for topographic surveys and map out the features with centimeter accuracy. It should be noted that for rugged terrain, it is preferred to increase the tie points between the photos, and a higher image overlap is preferred to compensate for the obstructed landmarks.

### **b. Air LiDAR**

Since the SD of DJI Zenmuse L1 Air LiDAR (“L1”) in horizontal and vertical positions are less than GreenValley Air LiDAR (“GreenValley”), this shows that L1 yields more precise results. However, the RMSE in Z values in L1 was much higher than GreenValley. This was likely caused by systematic error, thus a consistent height offset was applied and made the height measurements inaccurate.

In addition to the positional differences, the point cloud from L1 appeared to be denser than air GreenValley. This shows that L1 likely has collected more returns per unit area, providing finer detail and less pixel ambiguity, increasing the accuracy.

Overall, it is believed that L1 can be made more accurate than GreenValley, given the systematic error was corrected, its lower standard deviation shows the measurements were more precise and consistent.

## **Challenges Encountered**

There were multiple challenges encountered in all surveying steps.

Firstly, the dense tree canopy has obstructed many surveying areas, making it difficult to find suitable locations for additional check points. This was an environmental constraint, given that Tai Tong has many tree surroundings, and the terrain was covered by a dense canopy. This may have affected the surveying accuracy, due to having fewer potential check points, and fewer photos for remote and rugged areas.

Secondly, the surveying process for the GNSS control points and check points was not very streamlined. During the planning process for the points, we have designated 30 locations on the satellite image of survey area B. Unfortunately, as we arrived at the survey areas, many of the locations were obstructed by trees or had a small sky window. Even though it may look unobstructed and a wide sky window on the map, the locations were still subject to change. In a nutshell, this shows that planning can deviate from the results in practice, and we had to react fast and adapt to the environment so that we could change the control points and check points to a different location.

Thirdly, the initially generated 3D point cloud for the building was suboptimal. One of the possible reasons was that the photos were too close to the building, and the software may not be able to recognize and join the photos. Another reason could be the lack of the April tags in the oblique photos, due to being concentrated in the same area with few control points. We have added some tie points, directly from the common features on the buildings, and were able to generate a 3D point cloud using the oblique photos alone without using the nadir photos for the whole area. By tackling the most difficult part of the UAV survey, which was a detailed scan around a building, we could easily add the rest of the photos for survey area B for the full 3D model.

Fourthly, the control points might not be registered successfully in aerial triangulation, or the RMSE value of the calculation of these points might be big. The solution to this problem is to manually digitize control points, and to improve the accuracy of digitization to ensure that each point accurately points to the center of the actual control point. Also, it was divided into two parts to do the aerial triangulation, one is for the whole survey area, the other is for the building.

Lastly, the aerial triangulation (AT) yielded suboptimal results. After the initial run, many photos were not included in the point cloud and did not register to the targets. In the subsequent attempts, multiple tie points were added, yet a number of photos were still unused by the point cloud. In addition, the quality of the control points and tie points were also suboptimal. It took many attempts of adding tie points to register all of the photos. Overall, AT was time consuming and its results were not very predictable, posing a major challenge in data processing.

### **Self reflection and lessons learned**

We believe that UAV surveying can be complemented with Air LiDAR to achieve higher accuracy. While UAV surveying was efficient and can cover a large area, some areas might be blocked, and it was difficult to record the data, especially in shadowed areas. Sometimes switching channels, such as switching to intensity mode, was effective to recognize the ground physical features. Some areas were flown over multiple times, and the point cloud image was still blurry. Therefore, survey results can be improved by accompanying the UAV survey with air LiDAR to create a detailed 3D model.

We have also gained hands-on experience with flying the drones. Although the controls only involved 8 variables: ascend/descend, turn left/right, forward/backward, and leftward/rightward, it did take some time to get familiarized with the controls, especially since the left/right/forward/backward directions were in the drone's POV, not the pilot. We believe that this experience can train the pilot to gain some initial muscle memory. According to the Civil Aviation Department (2025), Category A2 and B SUAs require training. Therefore, initial training might help to gain permission to fly heavier drones for UAV surveying in the future.

Another aspect that we have learnt in this survey is the ability to solve problems. In field surveys, the environment was subject to many changes and uncertainties, such as the deviation of the final control and check point locations from the initial plan. Unlike in lecture or in the classroom, the answers usually relied on quick thinking, adapting to the changing environment, and teamwork. For example, one of the challenges was adapting to the environmental constraints. In survey area B, most of the areas were covered by trees, and the

location choices for control and check points were limited. We reassessed the site during the survey and made changes to the original plan. Ultimately, this flexibility and adaptation to the changes have allowed us to set up the control points with an improved sky window for GNSS accuracy and UAV recognition.

Furthermore, we also learned how important site selection is for both control points and tie points. Those points should be placed in open areas with a wide sky view and on the features that are distinctive and high contrast to ensure reliable detection. Also, the placement of April tags and tie point markers must also be planned carefully to keep clear of the vehicle paths and high-traffic footpaths to minimize the risk of damage or displacement. In addition, environmental factors should also be taken into account. For example, how wind or rain might impact marker stability and visibility. The factors mentioned above must be accounted for, or else, the accuracy will likely be negatively affected.

## **Recommendations**

Reviewing this field camp, our group may have several suggestions that could improve for future survey and overall learning experience.

### **Pre-camp preparation**

We strongly recommend providing a drone flight simulator in week 13. During our second day in the survey camp (the day for practicing our drone), we spent quite some time simply adjusting to the controls, which seemed quite pointless. The concept that “forward,” “backward,” “left,” and “right” are relative to the drone’s orientation and not its pilot’s position. Also, it can let us touch real objects to learn how to replace the battery of a drone, open its wings and controller, and gain an early understanding of how to connect to a mobile phone, etc., in class on week 13. This could reduce the amount of valuable camp time spent on just learning the basic skills.

### **Time Management**

The schedule of activities in the five-day camp was quite tight, especially taking time to practice and study the drone knowledge, settings, and rules for practical tests on the fourth day. Also on the final day, we had to submit the model, which was processed throughout the night. Due to time limitations, we did not have enough time to re-touch because the processing time of the reconstruction was lengthy. Therefore, I hope the camp can be extended to six days. Nevertheless, this experience reflects the essential characteristics of being a professional land surveyor for our future, as we must adhere to strict deadlines and ensure all work is submitted on time.

### **Parameter Settings**

For the parameter of setting the flight path of the UAV, we recommend that the overlap ratio be changed to 80% (originally 70%), since a higher overlap ratio makes it easier for the software to generate the model. Our survey area B contains a lot of terrain, including many trees. Trees may provide fewer tie points, which makes it difficult for the software to match common points. Therefore, increasing the percentage of overlap will make it easier for the software to form the model, reducing the time required for aerotriangulation.

## Mixed Drone Photography

We used both the Mavic Air 2 and Mini 5 Pro for modelling (which is first doing the oblique photo of the building, and the other for the flight path of the entire survey area). However, in iTwin Capture Modeler, a yellow warning notice appeared, explaining that photos from different physical cameras need to be placed in separate subdirectories for optimal processing. For similar tasks in the future, we should consider using the same drone for all photography. This can ensure optimal precision and performance, as the same physical camera can have an identical focal length and the same dimensions.

## Appendix

### A. References

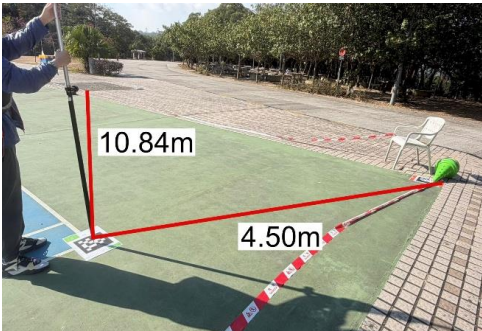
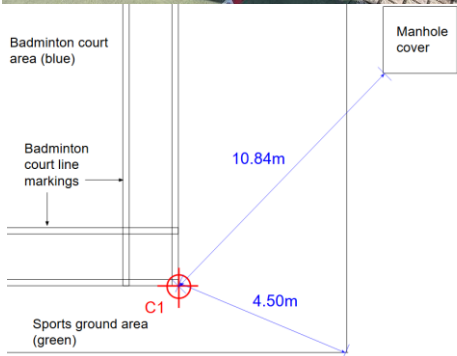
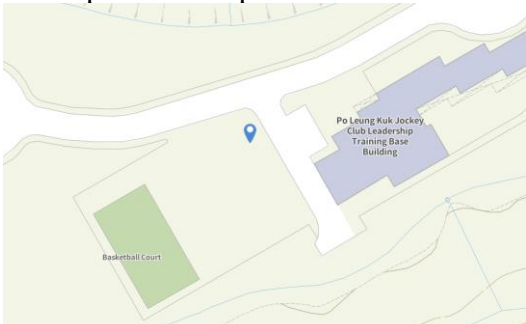
Civil Aviation Department (2025). *Small Unmanned Aircraft Order (Cap.448G)*.

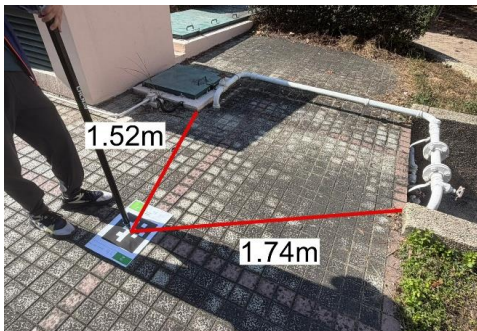
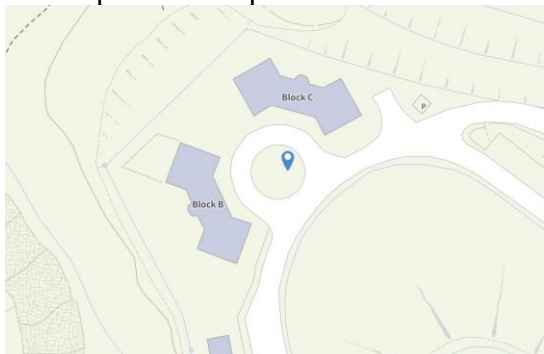
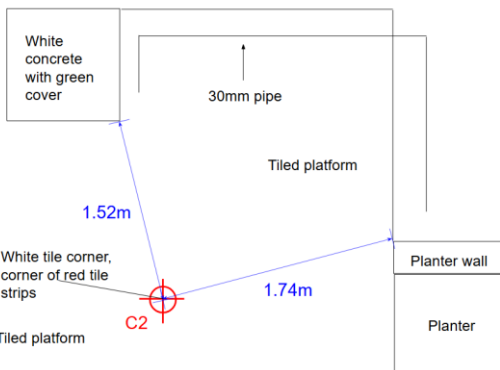
<https://www.cad.gov.hk/english/sua.html>

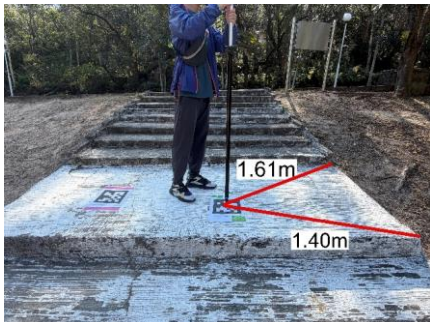
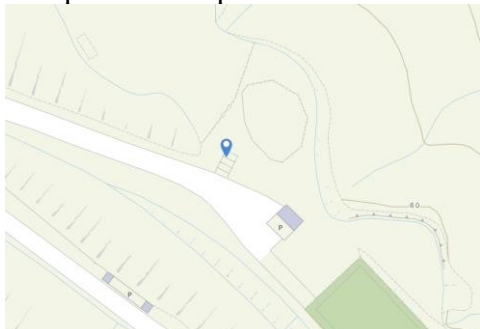
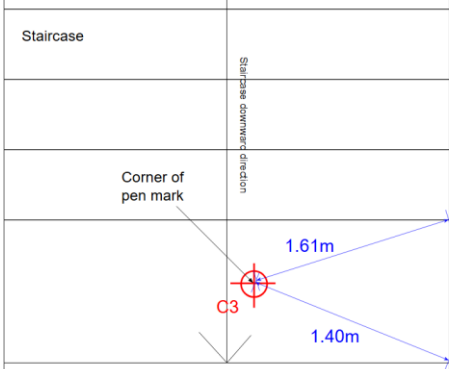
### B. GNSS RTK Results

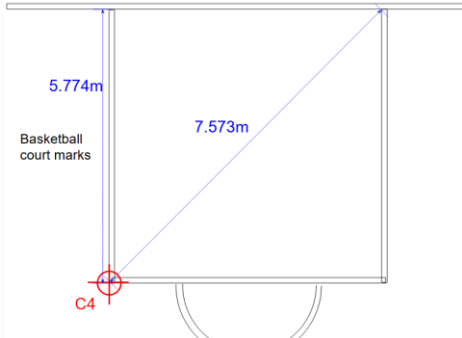
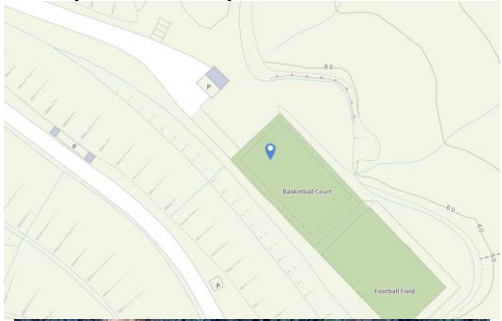
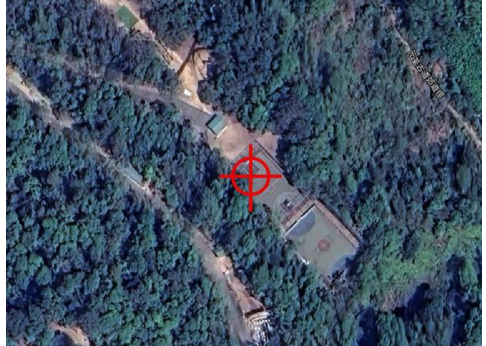
| Name | GNSS RTK   |            |        | CP2 Known Point |            |       | Difference (m) |       |       | Allowable ( $\pm 3$ cm to $\pm 5$ cm) |
|------|------------|------------|--------|-----------------|------------|-------|----------------|-------|-------|---------------------------------------|
|      | X          | Y          | Z      | X               | Y          | Z     | X              | Y     | Z     |                                       |
| CP2  | 821291.997 | 830233.929 | 69.017 | 821291.998      | 830233.935 | 69.05 | 0.001          | 0.006 | 0.033 | TRUE                                  |
| C1   | 821567.13  | 830138.748 | 91.408 |                 |            |       |                |       |       |                                       |
| T01  | 821569.072 | 830134.37  | 91.343 |                 |            |       |                |       |       |                                       |
| T02  | 821624.408 | 830163.271 | 91.401 |                 |            |       |                |       |       |                                       |
| T03  | 821631.015 | 830167.642 | 91.37  |                 |            |       |                |       |       |                                       |
| T04  | 821587.166 | 830153.109 | 91.043 |                 |            |       |                |       |       |                                       |
| T05  | 821579.589 | 830146.943 | 91.065 |                 |            |       |                |       |       |                                       |
| T06  | 821564.534 | 830137.204 | 91.429 |                 |            |       |                |       |       |                                       |
| C2   | 821578.628 | 830313.298 | 71.811 |                 |            |       |                |       |       |                                       |
| T07  | 821634.624 | 830305.018 | 70.842 |                 |            |       |                |       |       |                                       |
| T08  | 821674.139 | 830277.554 | 70.997 |                 |            |       |                |       |       |                                       |
| T09  | 821724.544 | 830218.86  | 70.453 |                 |            |       |                |       |       |                                       |
| T10  | 821710.79  | 830306.292 | 61.796 |                 |            |       |                |       |       |                                       |
| T11  | 821713.069 | 830297.904 | 62.758 |                 |            |       |                |       |       |                                       |
| T12  | 821716.616 | 830289.928 | 62.098 |                 |            |       |                |       |       |                                       |
| C3   | 821708.472 | 830305.872 | 62.274 |                 |            |       |                |       |       |                                       |
| C4   | 821737.628 | 830268.596 | 61.628 |                 |            |       |                |       |       |                                       |
| C5   | 821771.108 | 830240.738 | 61.44  |                 |            |       |                |       |       |                                       |
| T13  | 821764.096 | 830254.397 | 61.418 |                 |            |       |                |       |       |                                       |
| T14  | 821753.315 | 830244.445 | 61.515 |                 |            |       |                |       |       |                                       |
| T15  | 821749.059 | 830256.707 | 61.621 |                 |            |       |                |       |       |                                       |
| T16  | 821752.032 | 830259.594 | 61.572 |                 |            |       |                |       |       |                                       |
| T17  | 821746.321 | 830265.57  | 61.583 |                 |            |       |                |       |       |                                       |
| T18  | 821743.812 | 830263.133 | 61.617 |                 |            |       |                |       |       |                                       |
| T19  | 821741.181 | 830271.966 | 61.566 |                 |            |       |                |       |       |                                       |
| T20  | 821750.521 | 830269.471 | 61.539 |                 |            |       |                |       |       |                                       |
| C6   | 821727.402 | 830220.114 | 70.533 |                 |            |       |                |       |       |                                       |
| T21  | 821731.966 | 830220.379 | 70.765 |                 |            |       |                |       |       |                                       |
| CP2  | 821291.999 | 830233.928 | 69.015 | 821291.998      | 830233.935 | 69.05 | -0.001         | 0.007 | 0.035 | TRUE                                  |

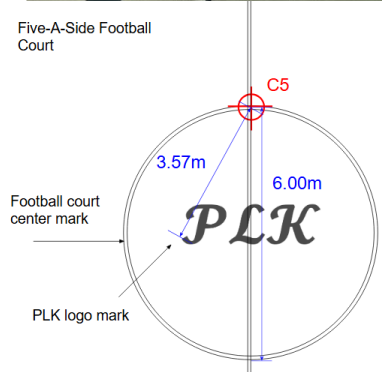
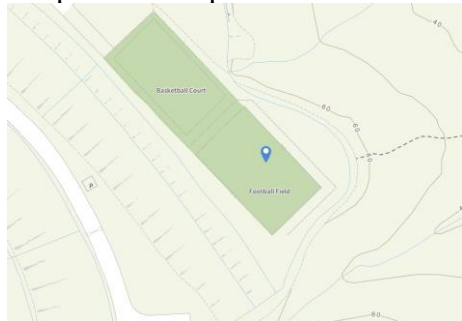
### C. Stations summaries for control points

|                                                                                                                                                                            |                                 |                                                                                                                                                                              |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Station name                                                                                                                                                               | C1                              | Easting                                                                                                                                                                      | 821567.130 |
| Description                                                                                                                                                                | Outer corner of badminton court | Northing                                                                                                                                                                     | 830138.748 |
|                                                                                                                                                                            |                                 | Height                                                                                                                                                                       | 91.408     |
| Sketch                                                                                                                                                                     |                                 | Control point on map                                                                                                                                                         |            |
| <br> |                                 | <br> |            |

|                                                                                    |                                        |                                                                                     |            |
|------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------|------------|
| Station name                                                                       | C2                                     | Easting                                                                             | 821578.628 |
| Description                                                                        | Tile corner, inner corner of red strip | Northing                                                                            | 830313.298 |
|                                                                                    |                                        | Height                                                                              | 71.811     |
| Sketch                                                                             |                                        | Control point on map                                                                |            |
|   |                                        |   |            |
|  |                                        |  |            |

|                                                                                     |                                 |                                                                                      |            |
|-------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------|------------|
| Station name                                                                        | C3                              | Easting                                                                              | 821708.472 |
| Description                                                                         | Corner of pen mark on staircase | Northing                                                                             | 830305.872 |
|                                                                                     |                                 | Height                                                                               | 62.274     |
| Sketch                                                                              |                                 | Control point on map                                                                 |            |
|  |                                 |  |            |
|  |                                 |  |            |

|                                                                                                                                                                      |                                           |                                                                                                                                                                        |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Station name                                                                                                                                                         | C4                                        | Easting                                                                                                                                                                | 821737.628 |
| Description                                                                                                                                                          | Corner of basketball court lane line mark | Northing                                                                                                                                                               | 830268.596 |
|                                                                                                                                                                      |                                           | Height                                                                                                                                                                 | 61.628     |
| Sketch                                                                                                                                                               |                                           | Control point on map                                                                                                                                                   |            |
|   |                                           |   |            |

|                                                                                                                                                                                                           |                                           |                                                                                                                                                                           |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Station name                                                                                                                                                                                              | C5                                        | Easting                                                                                                                                                                   | 821771.108 |
| Description                                                                                                                                                                                               | Corner of football court center ring mark | Northing                                                                                                                                                                  | 830240.738 |
|                                                                                                                                                                                                           |                                           | Height                                                                                                                                                                    | 61.440     |
| Sketch                                                                                                                                                                                                    |                                           | Control point on map                                                                                                                                                      |            |
|  <p>Five-A-Side Football Court</p>  |                                           |   |            |

|              |                                  |          |            |
|--------------|----------------------------------|----------|------------|
| Station name | C6                               | Easting  | 821727.402 |
| Description  | Corner of yellow arrow road mark | Northing | 830220.114 |
|              |                                  | Height   | 70.533     |

Sketch

Control point on map

D. Survey photos

T1

T2



T3



T4



T5



T6



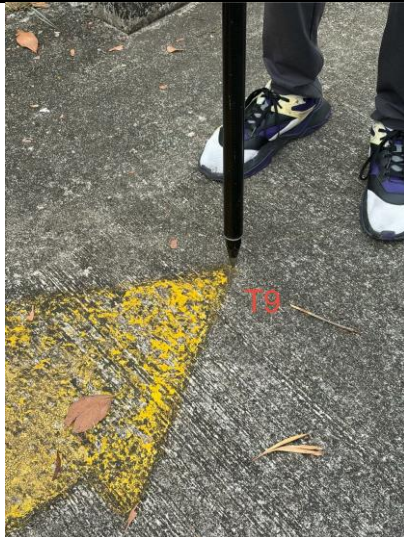
T7



T8



T9



T10



T11



T12



T13



T14



T15



T16



T17



T18



T19



T20



T21



### E. Production .3sm file screenshots





Building.3sm





Terrain.3sm

## F. Aerotriangulation quality reports

Quality reports were generated by iTwin to view the accuracy of the control points. Since 2 sets of 3D mesh models were generated for building and terrain respectively, 2 sets of reports were generated before final production and are shown below.

| Control Points Errors |          |                                     |                             |                                    |                                   |                   |                             |                         |   |
|-----------------------|----------|-------------------------------------|-----------------------------|------------------------------------|-----------------------------------|-------------------|-----------------------------|-------------------------|---|
| Name                  | Category | Accuracy [meters]                   | Number of Calibrated Photos | RMS of Reprojection Error [pixels] | RMS of Distances to Rays [meters] | 3D Error [meters] | Horizontal Error [meters]   | Vertical Error [meters] |   |
| T2                    | 3D       | Horizontal: 0.05;<br>Vertical: 0.05 | 4 (4 marked photos)         | 2.52                               | 0.0318                            | 0.03463           | X: 0.03311;<br>Y: -0.00989  | -0.00211                | ⚠ |
| T3                    | 3D       | Horizontal: 0.05;<br>Vertical: 0.05 | 6 (6 marked photos)         | 1.92                               | 0.01983                           | 0.02557           | X: -0.0243;<br>Y: 0.00729   | 0.00318                 | ⚠ |
| T4                    | 3D       | Horizontal: 0.05;<br>Vertical: 0.05 | 5 (5 marked photos)         | 0.36                               | 0.01772                           | 0.00154           | X: 0.00125;<br>Y: 0.00024   | -0.00086                | ✅ |
| T5                    | 3D       | Horizontal: 0.05;<br>Vertical: 0.05 | 5 (5 marked photos)         | 0.44                               | 0.01485                           | 0.00423           | X: 0.00232;<br>Y: 0.0035    | -0.0005                 | ✅ |
| Target #7             | 3D       | Horizontal: 0.01;<br>Vertical: 0.01 | 5 (5 marked photos)         | 0.44                               | 0.02614                           | 0.00537           | X: -0.00484;<br>Y: -0.00197 | 0.00126                 | ✅ |
| Global RMS            |          |                                     |                             | 1.46                               | 0.0229                            | 0.0195            | X: 0.01853;<br>Y: 0.00578   | 0.00185                 |   |
| Median                |          |                                     |                             | 0.44                               | 0.01983                           | 0.00537           | X: 0.00125;<br>Y: 0.00024   | -0.0005                 |   |

### ▲ Building.3sm control points errors

| User Tie Points Errors |                             |                                    |                                   |   |
|------------------------|-----------------------------|------------------------------------|-----------------------------------|---|
| Name                   | Number of Calibrated Photos | RMS of Reprojection Error [pixels] | RMS of Distances to Rays [meters] |   |
| Target #14             | 1 (1 marked photos)         |                                    |                                   |   |
| Target #37             | 3 (3 marked photos)         | 0.32                               | 0.00342                           | ✓ |
| Target #46             | 10 (10 marked photos)       | 0.93                               | 0.00531                           | ✓ |
| User Tie Point 1       | 5 (5 marked photos)         | 1.36                               | 0.01163                           | ⚠ |
| User Tie Point 2       | 4 (4 marked photos)         | 1.91                               | 0.01355                           | ⚠ |
| User Tie Point 3       | 4 (4 marked photos)         | 0.55                               | 0.0159                            | ✓ |
| User Tie Point 4       | 4 (5 marked photos)         | 0.64                               | 0.0074                            | ✓ |
| User Tie Point 5       | 4 (5 marked photos)         | 0.56                               | 0.00512                           | ✓ |
| User Tie Point 6       | 3 (4 marked photos)         | 0.46                               | 0.00469                           | ✓ |
| User Tie Point 7       | 3 (3 marked photos)         | 0.46                               | 0.00396                           | ✓ |
| User Tie Point 8       | 3 (4 marked photos)         | 0.74                               | 0.01278                           | ✓ |
| User Tie Point 9       | 3 (4 marked photos)         | 0.83                               | 0.00994                           | ✓ |
| User Tie Point 10      | 2 (4 marked photos)         | 0.03                               | 0.00804                           | ✓ |
| User Tie Point 11      | 4 (4 marked photos)         | 1.2                                | 0.00869                           | ⚠ |
| User Tie Point 12      | 3 (5 marked photos)         | 0.48                               | 0.00989                           | ✓ |
| User Tie Point 13      | 4 (6 marked photos)         | 1.7                                | 0.01671                           | ⚠ |
| User Tie Point 14      | 3 (6 marked photos)         | 0.39                               | 0.0034                            | ✓ |
| User Tie Point 15      | 3 (4 marked photos)         | 0.36                               | 0.01566                           | ✓ |
| Global RMS             |                             | 0.91                               | 0.01019                           |   |
| Median                 |                             | 0.56                               | 0.00869                           |   |

▲ *Building.3sm user tie points errors*

| Control Points Errors |          |                                  |                             |                                    |                                   |                   |                           |                         |   |
|-----------------------|----------|----------------------------------|-----------------------------|------------------------------------|-----------------------------------|-------------------|---------------------------|-------------------------|---|
| Name                  | Category | Accuracy [meters]                | Number of Calibrated Photos | RMS of Reprojection Error [pixels] | RMS of Distances to Rays [meters] | 3D Error [meters] | Horizontal Error [meters] | Vertical Error [meters] |   |
| Target                | 3D       | Horizontal: 0.05;                | 16 (16                      | 2.74                               | 0.03345                           | 0.03231           | X: -0.02478;              | -0.00403                | ⚠ |
| #7                    |          | Vertical: 0.05                   | marked photos)              |                                    |                                   |                   | Y: -0.02035               |                         |   |
| Target #8             | 3D       | Horizontal: 0.05; Vertical: 0.05 | 8 (8 marked photos)         | 1.89                               | 0.0425                            | 0.052             | X: 0.04207; Y: -0.02428   | 0.01857                 | ⚠ |
| Target #9             | 3D       | Horizontal: 0.05; Vertical: 0.05 | 15 (15 marked photos)       | 1.33                               | 0.02526                           | 0.03095           | X: 0.0158; Y: 0.01246     | -0.02352                | ⚠ |
| Target #10            | 3D       | Horizontal: 0.05; Vertical: 0.05 | 15 (15 marked photos)       | 2.09                               | 0.02559                           | 0.03032           | X: 0.00384; Y: 0.02111    | -0.02142                | ⚠ |
| Target #11            | 3D       | Horizontal: 0.05; Vertical: 0.05 | 17 (17 marked photos)       | 2.61                               | 0.02028                           | 0.02204           | X: 0.00552; Y: 0.01708    | 0.01279                 | ⚠ |
| Target #12            | 3D       | Horizontal: 0.05; Vertical: 0.05 | 14 (14 marked photos)       | 1.54                               | 0.03632                           | 0.04738           | X: -0.03966; Y: -0.00656  | 0.02506                 | ⚠ |
| Global RMS            |          |                                  |                             | 2.10                               | 0.03149                           | 0.03732           | X: 0.02662; Y: 0.01798    | 0.01899                 |   |
| Median                |          |                                  |                             | 2.09                               | 0.03345                           | 0.03231           | X: 0.00552; Y: 0.01246    | 0.01279                 |   |

▲ *Terrain.3sm control points errors*

|                   |                     |       |         |   |
|-------------------|---------------------|-------|---------|---|
| User Tie Point 3  | 4 (4 marked photos) | 0.97  | 0.02137 | ✓ |
| User Tie Point 4  | 4 (4 marked photos) | 0.81  | 0.01828 | ✓ |
| User Tie Point 5  | 4 (4 marked photos) | 0.48  | 0.0108  | ✓ |
| User Tie Point 6  | 4 (4 marked photos) | 0.75  | 0.01729 | ✓ |
| Target #3         | 5 (5 marked photos) | 0.59  | 0.00212 | ✓ |
| Target #26        | 2 (2 marked photos) | 0.34  | 0.00145 | ✓ |
| Target #36        | 4 (4 marked photos) | 0.76  | 0.00327 | ✓ |
| Target #37        | 2 (2 marked photos) | 0.35  | 0.00211 | ✓ |
| Target #40        | 5 (5 marked photos) | 0.67  | 0.00295 | ✓ |
| Target #41        | 4 (4 marked photos) | 0.56  | 0.0025  | ✓ |
| Target #6         | 4 (4 marked photos) | 0.31  | 0.00212 | ✓ |
| Target #57        | 4 (4 marked photos) | 0.63  | 0.0047  | ✓ |
| Target #58        | 2 (2 marked photos) | 0.6   | 0.00133 | ✓ |
| User Tie Point 7  | 4 (4 marked photos) | 0.35  | 0.00773 | ✓ |
| User Tie Point 8  | 4 (4 marked photos) | 1.24  | 0.02865 | ⚠ |
| User Tie Point 9  | 4 (4 marked photos) | 0.45  | 0.010   | ✓ |
| User Tie Point 10 | 4 (4 marked photos) | 0.79  | 0.01617 | ✓ |
|                   |                     |       |         | ✓ |
| User Tie Point 11 | 4 (4 marked photos) | 0.88  | 0.01989 | ✓ |
| User Tie Point 12 | 4 (4 marked photos) | 0.77  | 0.01469 | ✓ |
| Global RMS        |                     | 36.45 | 0.50067 |   |
| Median            |                     | 0.67  | 0.010   |   |

▲ *Terrain.3sm user tie points errors*